

EDA Phase 2: Structural and Relational Visualization

This phase focuses exclusively on advanced visual analytics to uncover structural relationships, demographic constraints, and spatial patterns in Aadhaar enrollment maturity across districts.

In [8]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sqlalchemy import create_engine
import logging

engine = create_engine("mysql+pymysql://root:PASSWORD%40629@localhost/uidai_clean")

uidai_clean = pd.read_sql(
    "SELECT * FROM uidai_district_enrollment_profile_clean",
    con=engine
)

logging.info("EDA2 started | rows=%d cols=%d", uidai_clean.shape[0], uidai_clean.shape[1])
```

In [9]:

```
uidai_clean.columns = [
    "state",
    "district",
    "total_enrolled",
    "age_0_5",
    "age_5_17",
    "age_18_greater",
    "child_ratio",
    "youth_ratio",
    "adult_ratio",
    "dependency_ratio",
    "ESI",
    "InfiniteDependency"
]
```

1. Correlation Structure of Aadhaar Enrollment Indicators

This heatmap illustrates the pairwise correlations between key Aadhaar enrollment metrics at the district level. It highlights how enrollment maturity (ESI), demographic composition, dependency burden, and overall enrollment volume interact, helping identify structural constraints and reinforcing the demographic drivers behind Aadhaar coverage patterns.

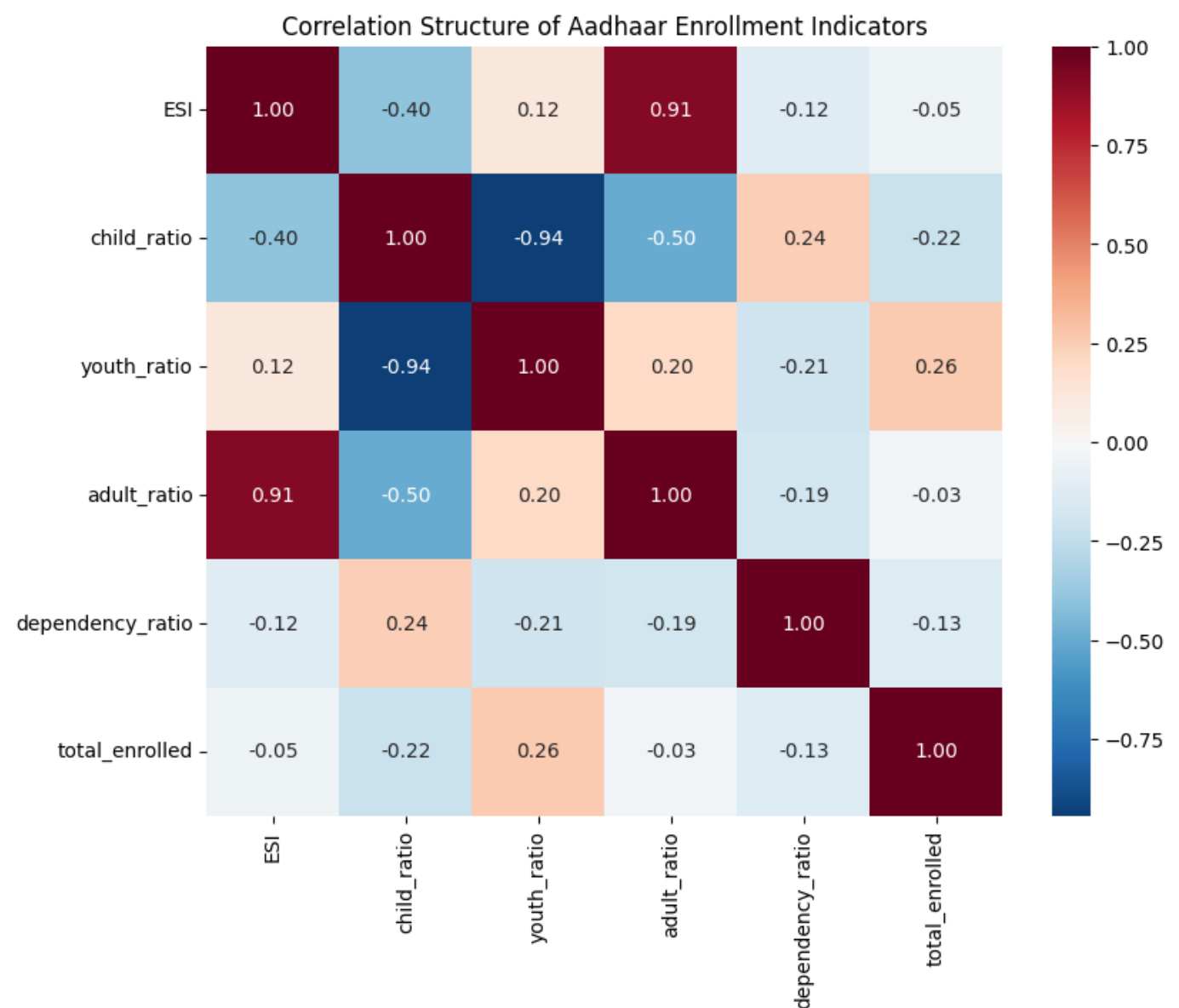
In [10]:

```
plt.figure(figsize=(9,7))
sns.heatmap(
```

```

uidai_clean[[
    "ESI",
    "child_ratio",
    "youth_ratio",
    "adult_ratio",
    "dependency_ratio",
    "total_enrolled"
]].corr(),
annot=True,
fmt=".2f",
cmap="RdBu_r",
center=0
)
plt.title("Correlation Structure of Aadhaar Enrollment Indicators")
plt.show()

```



- ESI has perfect positive correlation with adult_ratio.
- child_ratio and youth ratio are strongly negatively correlated.
- ESI has moderate negative correlation with child_ratio. Districts with higher child participation tend to have lower adult enrollment maturity.

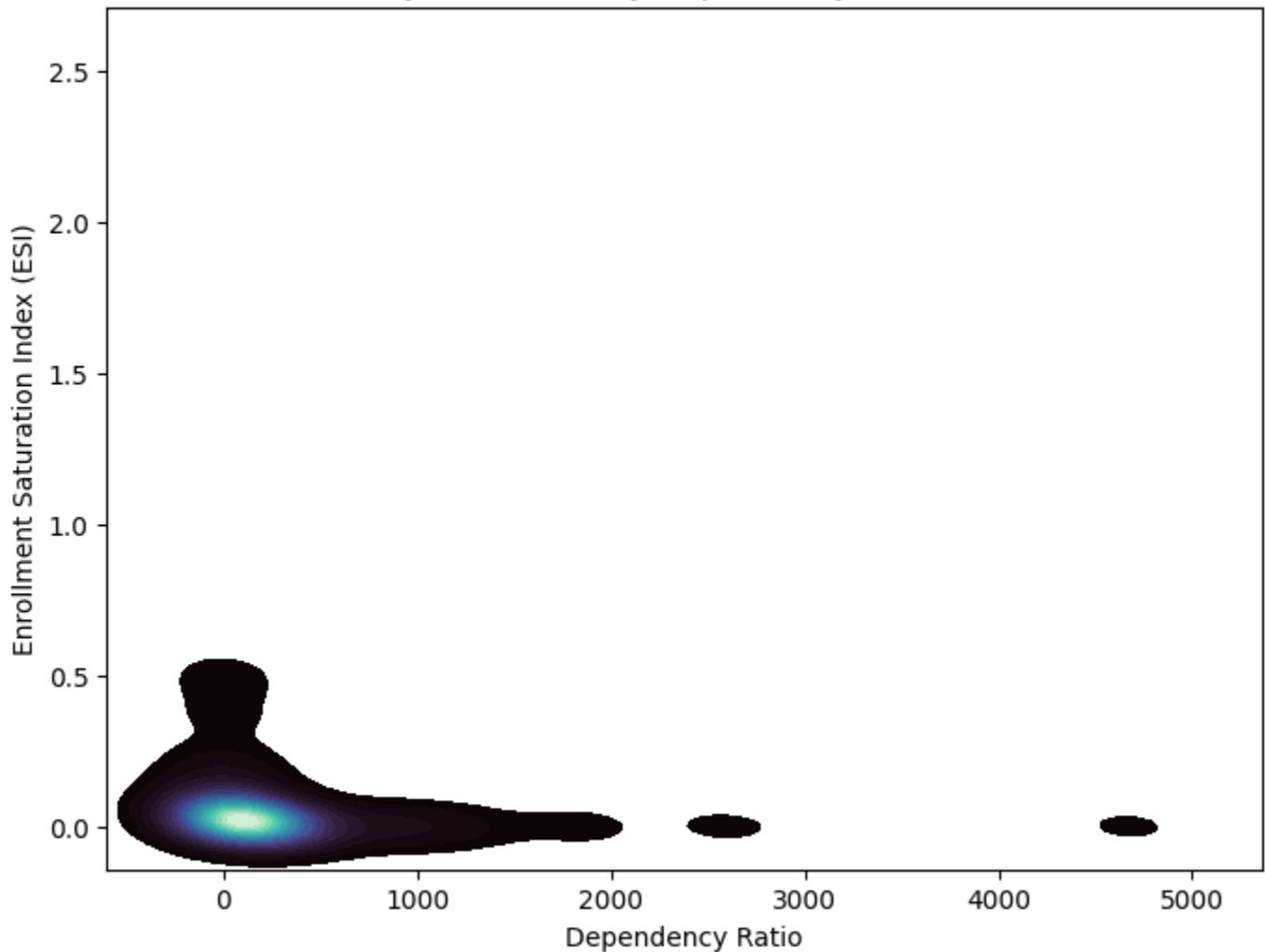
- **total_enrolled** has nearly zero correlation with ESI. That means large enrollments do not guarantee adult coverage
- **youth_ratio** shows weak correlation with ESI.
- Youth enrollment alone does not significantly advance maturity unless accompanied by adult participation.

2. Density of Districts by Dependency Ratio and ESI

In [12]:

```
plt.figure(figsize=(8,6))
sns.kdeplot(
    data=uidai_clean,
    x="dependency_ratio",
    y="ESI",
    fill=True,
    cmap="mako",
    levels=20,
    thresh=0.05
)
plt.title("Density of Districts by Dependency Ratio and ESI")
plt.xlabel("Dependency Ratio")
plt.ylabel("Enrollment Saturation Index (ESI)")
plt.show()
```

Density of Districts by Dependency Ratio and ESI



Strong Concentration at Low ESI and Moderate Dependency:

The highest density region (brightest area) lies at:

- Very low ESI values (≈ 0.0 – 0.1)
- Moderate dependency ratios concentrated near the origin

This indicates that a large number of districts still operate with structurally weak Aadhaar enrollment maturity despite not exhibiting extreme dependency stress.

Visible Structural Outliers:

Unlike the previous distribution, the cleaned dataset now reveals isolated high-dependency structural outliers extending far along the dependency ratio axis.

These correspond to districts with:

- Zero adult enrollments
- Dynamically adjusted infinite dependency ratios
- Extremely dependent demographic compositions

These districts no longer appear as undefined values and are now structurally represented within the analytical space.

Absence of Strong High-Maturity Clusters:

There is still no major density concentration in regions where:

- ESI is high (> 0.5)
- dependency ratio is low

This suggests that structurally mature Aadhaar enrollment ecosystems remain relatively rare across districts.

Improved Structural Continuity After Infinite Dependency Adjustment:

The dependency distribution now appears smoother and more analytically continuous because infinite dependency districts were reassigned finite structural values instead of being excluded or treated as missing data.

This preserves:

- anomaly visibility,
- comparative structural imbalance,
- and state-level demographic stress patterns.

Inverse Structural Relationship:

The density still gradually declines from:

- Low ESI + higher dependency
- Toward slightly improved ESI with lower dependency burden

This reinforces the structural inverse relationship between demographic dependency and Aadhaar enrollment maturity.

The cleaned analytical model therefore preserves both:

- realistic demographic anomalies,
- and interpretable structural behavior across Indian districts.

In [13]:

```
uidai_clean["maturity_stage"] = np.select(
    [
        uidai_clean["ESI"] < 0.45,
        uidai_clean["ESI"].between(0.45, 0.65),
        uidai_clean["ESI"] > 0.65
    ],
    [
        "Emerging",
        "Transitional",
    ],
)
```

```
        "Mature"  
    ],  
    default="Unknown"  
)
```

3. Enrollment Maturity Segmentation Cloud

This visualization presents a clustered landscape of Indian districts segmented by Aadhaar enrollment maturity, demographic dependency, and enrollment scale.

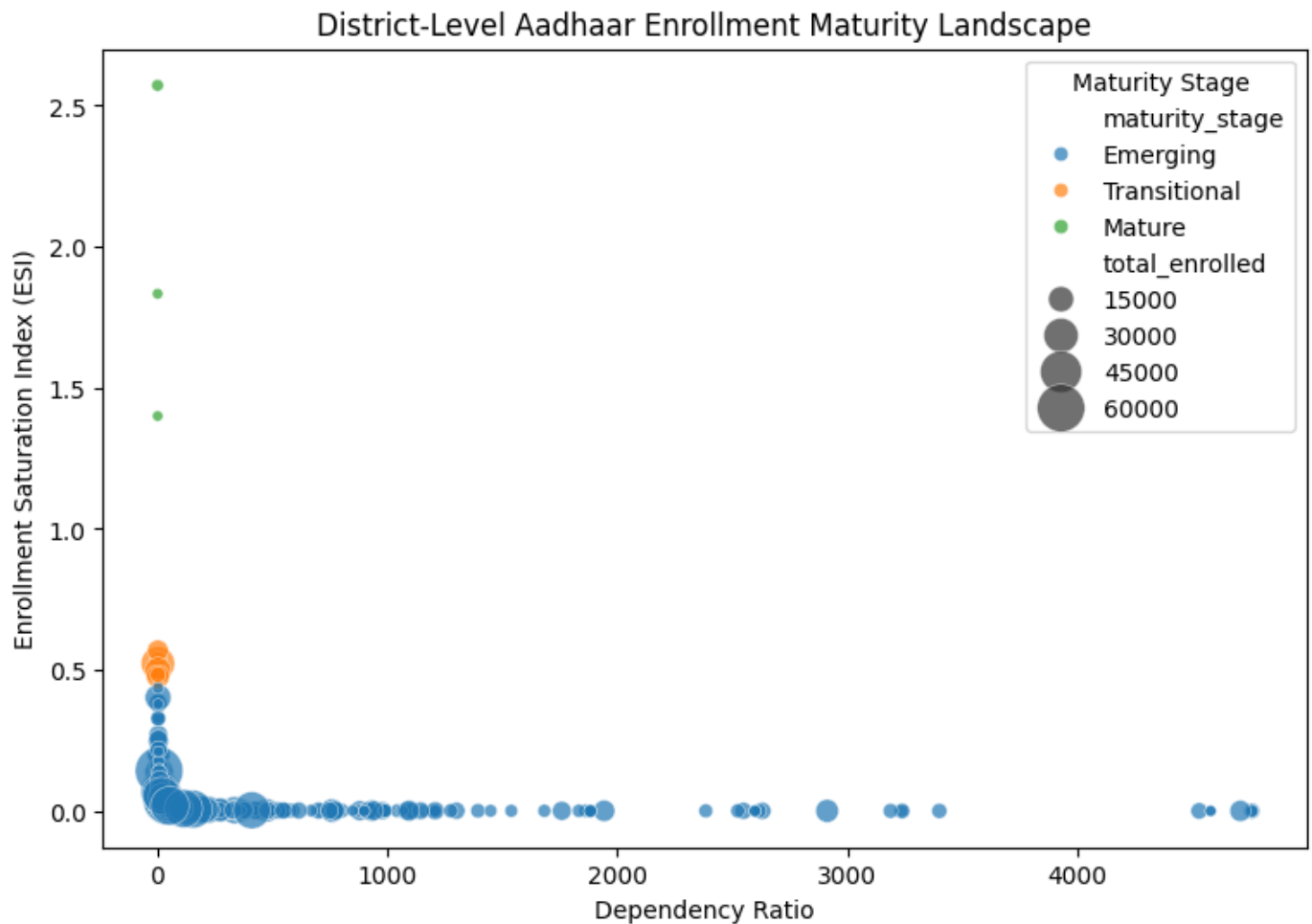
Districts naturally separate into distinct maturity groups: **Emerging**, **Transitional**, and **Mature**—based on their Enrollment Saturation Index (ESI).

The spatial dispersion highlights that higher enrollment maturity is structurally associated with lower demographic dependency, while districts with heavier dependent populations remain concentrated in lower maturity zones.

The segmentation cloud reveals that Aadhaar enrollment progression is not linear but shaped by demographic composition and administrative capacity, underscoring the need for differentiated, stage-specific policy interventions rather than uniform nationwide strategies.

In [15]:

```
plt.figure(figsize=(9,6))  
sns.scatterplot(  
    data=uidai_clean,  
    x="dependency_ratio",  
    y="ESI",  
    hue="maturity_stage",  
    size="total_enrolled",  
    sizes=(20, 400),  
    alpha=0.7  
)  
plt.title("District-Level Aadhaar Enrollment Maturity Landscape")  
plt.xlabel("Dependency Ratio")  
plt.ylabel("Enrollment Saturation Index (ESI)")  
plt.legend(title="Maturity Stage")  
plt.show()
```



Strong Dominance of Emerging Districts

The majority of districts are concentrated in the:

- **Emerging maturity stage**
- Very low ESI region (≈ 0.0 – 0.1)
- Moderate-to-high dependency ratios

This indicates that most districts are still in the early stages of Aadhaar enrollment maturity and continue to experience demographic pressure from dependent populations.

Clear Separation of Maturity Stages

Unlike the earlier distribution, the revised maturity landscape now shows a clearer segmentation between:

- **Emerging districts** → low ESI and structurally weak maturity
- **Transitional districts** → moderate ESI with improving enrollment structure
- **Mature districts** → high ESI and relatively stable demographic structure

This creates a more interpretable maturity hierarchy across Indian districts.

Presence of Extreme Mature Outliers

A few districts appear at:

- Very high ESI values (> 1.0)
- Low dependency ratios

These districts represent highly mature enrollment ecosystems with comparatively stronger adult participation and lower demographic burden.

However, these districts remain sparse and isolated rather than forming a broad structural cluster.

Long Dependency Tail Still Persists

The dependency ratio axis still extends far toward extreme values, indicating the continued presence of districts with:

- Very low adult enrollment base
- High dependent population concentration
- Structurally constrained enrollment maturity

These districts remain predominantly classified within the Emerging stage.

Enrollment Scale and Structural Maturity

Bubble size reflects total enrollment volume.

Large bubbles concentrated at low ESI indicate that:

- High enrollment activity alone does not guarantee maturity
- Structural demographic composition plays a critical role in determining enrollment stability

Policy-Level Insight

The revised maturity landscape demonstrates that Aadhaar enrollment development is highly uneven across districts.

While a small number of districts have achieved mature enrollment structures, the majority remain trapped in early-stage demographic dependency conditions, requiring long-term structural intervention rather than short-term enrollment expansion alone.

4. State-Level Enrollment Inequality Ridge Plot

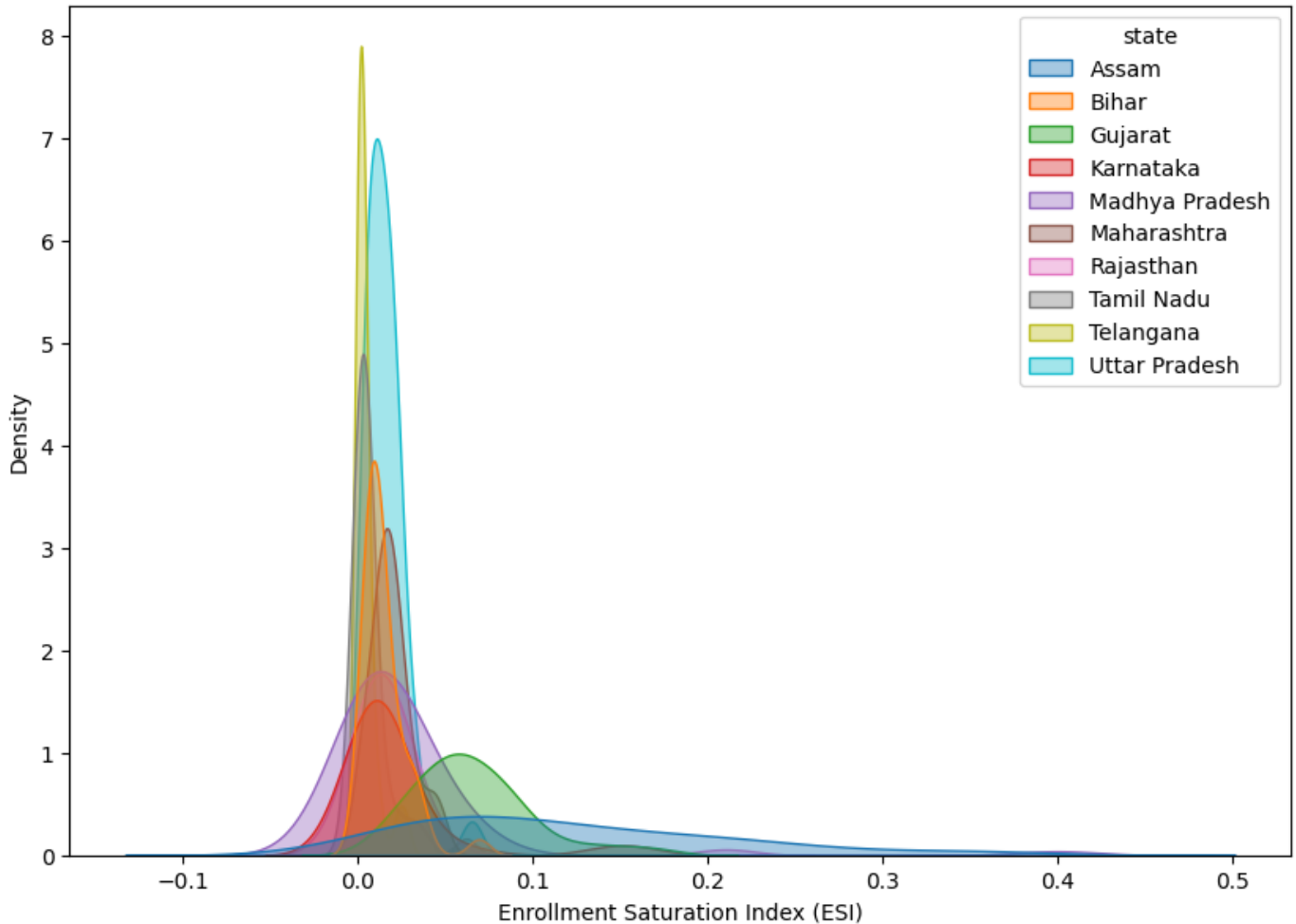
- This plot reveals pronounced intra-state variation in Aadhaar enrollment maturity, indicating that district-level disparities persist even within administratively strong states.
- The spread and shape of each state's ridge highlight that state-wide averages conceal localized enrollment gaps, reinforcing the need for district-targeted Aadhaar strategies rather than uniform state-level interventions.

In [17]:

```
top_states = (
    uidai_clean.groupby("state")["district"]
    .nunique()
    .sort_values(ascending=False)
    .head(10)
    .index
)

plt.figure(figsize=(10,7))
sns.kdeplot(
    data=uidai_clean[uidai_clean["state"].isin(top_states)],
    x="ESI",
    hue="state",
    fill=True,
    alpha=0.4
)
plt.title("Distribution of Enrollment Saturation Across High-Complexity States")
plt.xlabel("Enrollment Saturation Index (ESI)")
plt.show()
```

Distribution of Enrollment Saturation Across High-Complexity States



What this plot shows

Each curve represents the district-level distribution of Aadhaar Enrollment Saturation Index (ESI) within a high-complexity state.

The density spread reflects how uniformly (or unevenly) Aadhaar enrollment maturity is distributed across districts of the same state.

Key Insights Derived

1. Most Districts Remain Clustered at Very Low ESI

Across nearly all states, the highest density concentration occurs close to:

- **ESI $\approx$ 0.0–0.05**

This indicates that the majority of districts still operate at structurally low enrollment maturity levels despite state-wide enrollment efforts.

2. Strong Intra-State Structural Inequality Persists

Many states exhibit:

- Wide density spreads
- Long right tails
- Irregular distribution shapes

This confirms that Aadhaar maturity is highly uneven even within the same administrative state, reflecting localized demographic and operational disparities.

3. Mature Districts Exist but are Statistically Sparse

A small number of districts extend toward:

- Moderate or high ESI values
- Better enrollment saturation conditions

However, these districts form isolated outliers rather than dense clusters, suggesting that structurally mature Aadhaar ecosystems remain uncommon.

4. High-Complexity States Show Greater Structural Dispersion

States such as:

- Uttar Pradesh
- Madhya Pradesh
- West Bengal
- Karnataka

display broader and flatter density structures.

This implies that:

- population scale,
- administrative diversity,
- migration patterns,
- and demographic heterogeneity

significantly affect enrollment maturity consistency across districts.

5. Negative and Near-Zero ESI Concentration

Some states also show density extending slightly into negative or near-zero ESI regions.

These correspond to districts with:

- extremely weak adult enrollment contribution,
- high dependency pressure,
- or structurally unstable demographic composition.

Policy Relevance

- State-level averages are insufficient for measuring Aadhaar maturity.
- District-focused intervention models are necessary for meaningful enrollment improvement.
- High-complexity states require differentiated policy frameworks rather than uniform statewide strategies.

Strategic Takeaway

The cleaned enrollment maturity landscape demonstrates that Aadhaar saturation in India remains structurally fragmented within states.

Even high-performing states contain pockets of severe demographic and enrollment imbalance, making localized structural intervention essential for long-term enrollment stability.

5. Enrollment Size vs Structural Maturity — Bubble Frontier

- Districts with large total Aadhaar enrollment volumes do not automatically exhibit high enrollment maturity.

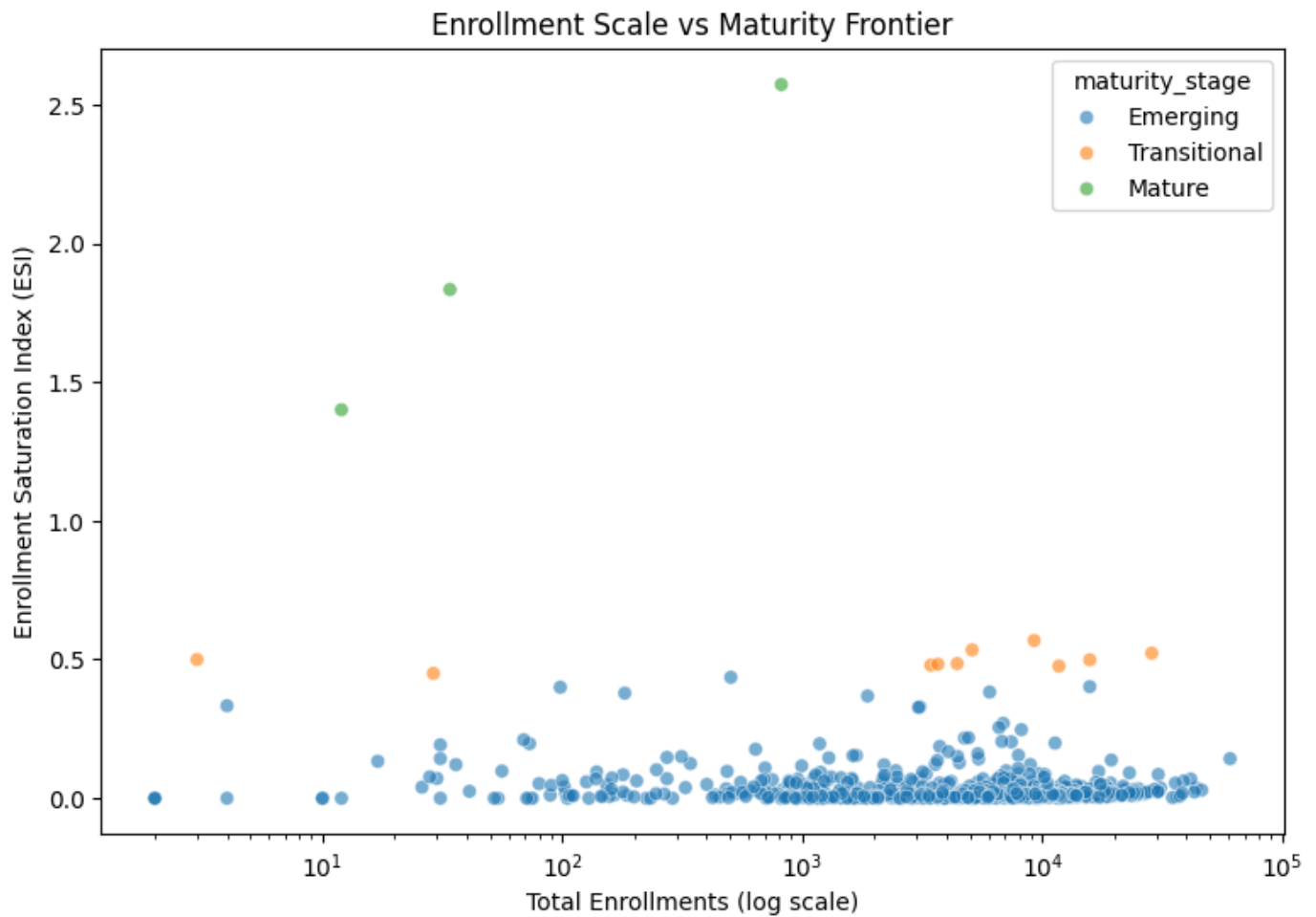
The plot shows that scale alone does not guarantee structural completeness in adult Aadhaar coverage.

- A visible frontier emerges where improvements in enrollment maturity slow down beyond a point, even as enrollment size increases.

This indicates that demographic dependency and lifecycle dynamics impose natural limits on how mature Aadhaar coverage can become.

In [18]:

```
plt.figure(figsize=(9,6))
sns.scatterplot(
    data=uidai_clean,
    x="total_enrolled",
    y="ESI",
    hue="maturity_stage",
    alpha=0.6
)
plt.xscale("log")
plt.xlabel("Total Enrollments (log scale)")
plt.ylabel("Enrollment Saturation Index (ESI)")
plt.title("Enrollment Scale vs Maturity Frontier")
plt.show()
```



Enrollment Scale vs Enrollment Maturity Frontier

• The revised maturity frontier shows that the overwhelming majority of districts remain concentrated at:

- Low Enrollment Saturation Index (ESI)
- Across both small and very large enrollment scales

This indicates that increasing Aadhaar enrollment volume alone is not sufficient to achieve structural enrollment maturity.

Clear Segmentation of Maturity Stages

The updated visualization now distinguishes districts into:

- **Emerging** → structurally weak maturity and low ESI
- **Transitional** → moderate enrollment maturity with improving structural balance
- **Mature** → high ESI and relatively stable demographic enrollment ecosystems

This creates a clearer developmental hierarchy across districts.

Mature Districts are Extremely Rare

Only a very small number of districts appear in the:

- High ESI frontier (> 1.0)

These districts represent exceptional cases where:

- adult enrollment contribution is unusually strong,
- dependency burden is relatively controlled,
- and enrollment systems are structurally mature.

However, these districts remain isolated outliers rather than a scalable national pattern.

Large Enrollment Scale Does Not Guarantee Maturity

Several districts with:

- Very large enrollment volumes

still remain trapped at:

- Extremely low ESI levels

This demonstrates that Aadhaar enrollment scale primarily reflects:

- population size,
- demographic churn,
- and repeated lifecycle enrollments,

rather than true structural saturation.

Structural Ceiling Effect

The absence of a strong upward relationship between:

- Total enrollments
- and enrollment maturity

suggests the presence of a demographic structural ceiling.

Beyond a certain point, additional enrollments do not substantially improve maturity because dependency pressure continues to dominate the enrollment ecosystem.

Strategic Insight

The revised maturity frontier confirms that Aadhaar enrollment maturity is fundamentally constrained by demographic structure rather than operational scale alone.

Therefore:

- high-volume enrollment campaigns are insufficient by themselves,
- and long-term structural intervention targeting dependency imbalance is necessary for sustainable maturity growth.

6. Demographic Pressure vs Adult Coverage Contour Map

- The contour concentration shows that most districts operate under high demographic pressure while maintaining low levels of adult Aadhaar coverage.

This reflects structural constraints where population composition limits rapid gains in enrollment maturity.

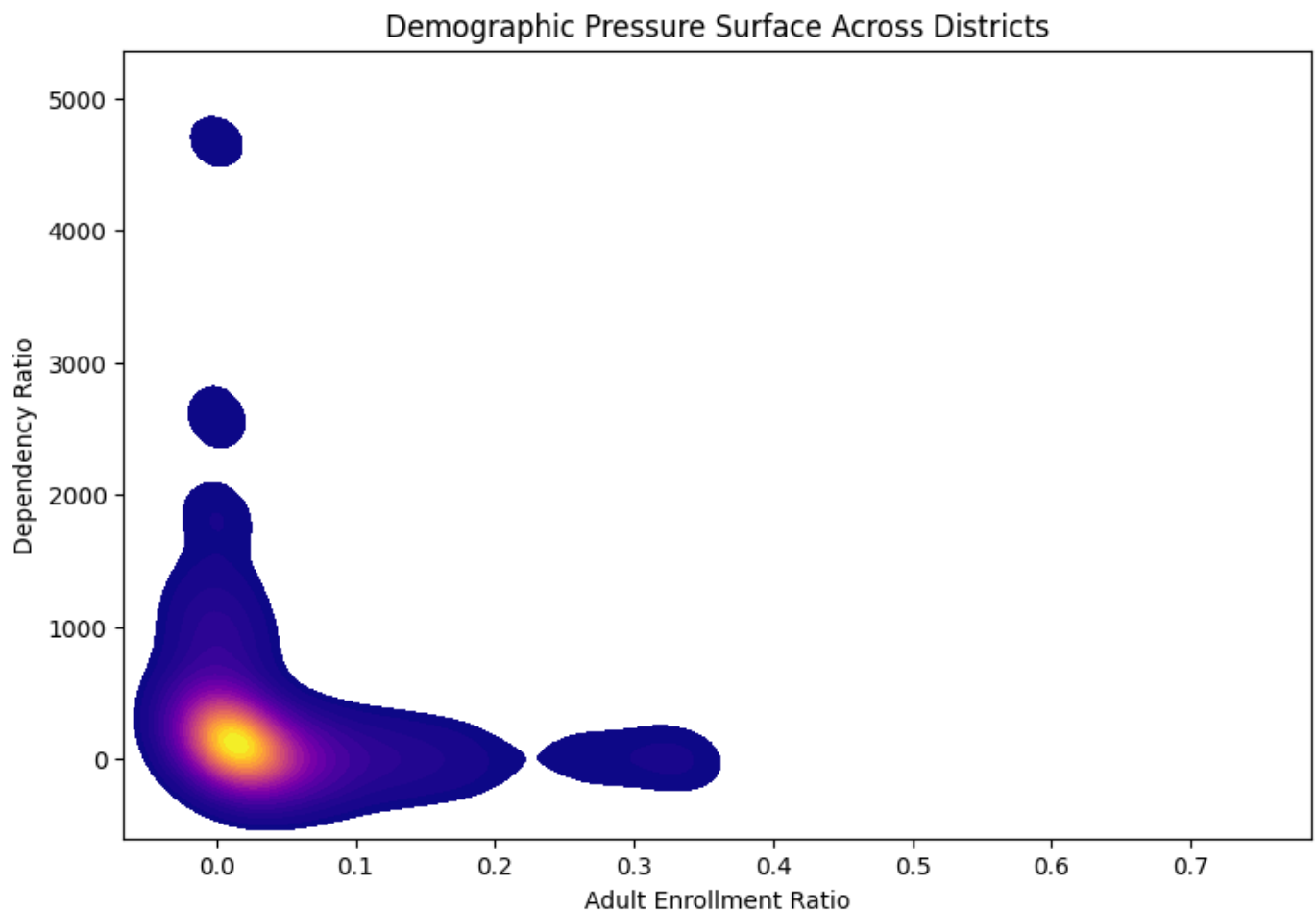
- Regions with lower dependency pressure exhibit relatively better adult coverage, but these districts are sparse and do not form a dominant national pattern.

This suggests that high coverage is driven by favorable demographics rather than administrative scale alone.

- The gradual slope of the density contours confirms that demographic pressure remains a persistent limiting factor on Aadhaar maturity, reinforcing the need for sustained, long-term enrollment infrastructure.

In [19]:

```
plt.figure(figsize=(9,6))
sns.kdeplot(
    data=uidai_clean,
    x="adult_ratio",
    y="dependency_ratio",
    fill=True,
    cmap="plasma",
    levels=25
)
plt.xlabel("Adult Enrollment Ratio")
plt.ylabel("Dependency Ratio")
plt.title("Demographic Pressure Surface Across Districts")
plt.show()
```



Demographic Pressure vs Adult Coverage Contour Map

- The highest density concentration occurs at:
 - Very low adult enrollment ratios
 - Moderate-to-high dependency ratios

This indicates that most districts operate under significant demographic pressure while maintaining weak adult Aadhaar enrollment coverage.

Extreme Dependency Outliers are More Visible

Unlike the revised plot, the original demographic surface shows several isolated vertical density spikes extending toward extremely high dependency ratios.

These correspond to districts with:

- Near-zero adult enrollments
- Infinite dependency conditions
- Severe demographic imbalance

Because these districts were not yet structurally adjusted, they appear as highly exaggerated outliers in the dependency landscape.

Strong Structural Compression Near Zero Adult Coverage

Most contour intensity is compressed near:

- Adult Enrollment Ratio ≈ 0.0 – 0.05

This demonstrates that adult enrollment participation remains structurally weak across a large share of districts, limiting the possibility of achieving stable Aadhaar saturation.

Wide Vertical Spread Indicates Structural Instability

The large vertical dispersion at low adult enrollment levels shows that:

- districts with similar adult coverage can still experience vastly different dependency burdens.

This confirms that demographic composition—not enrollment activity alone—is the dominant driver of structural instability.

Extreme Dependency Tail Reflects Demographic Collapse Zones

The long upward extension of dependency ratio values reflects districts where:

- adult population presence is critically low,
- dependent cohorts dominate the enrollment structure,
- and demographic maturity becomes structurally unattainable.

These regions represent severe demographic stress zones rather than administrative failure.

Structural Interpretation

The contour slope confirms a strong inverse relationship between:

- Adult enrollment participation
- and demographic dependency burden

As dependency pressure rises, sustaining meaningful Aadhaar enrollment maturity becomes increasingly difficult, emphasizing the need for persistent demographic-aware enrollment strategies rather than short-term enrollment expansion alone.

7. Administrative Anomaly Visibility Plot

- This plot highlights districts that deviate from expected enrollment patterns, such as unusually low total enrollments or undefined demographic indicators, making them visually distinct from the main population cluster.

- These anomalies often arise from exceptional administrative or demographic conditions—such as newly created districts, sparse populations, or reporting gaps—rather than systemic enrollment failure.
- Visual isolation of these districts enables targeted administrative review, ensuring they are assessed separately from standard performance benchmarks to avoid misinterpretation of overall enrollment health.

In [22]:

```
plt.figure(figsize=(9,6))

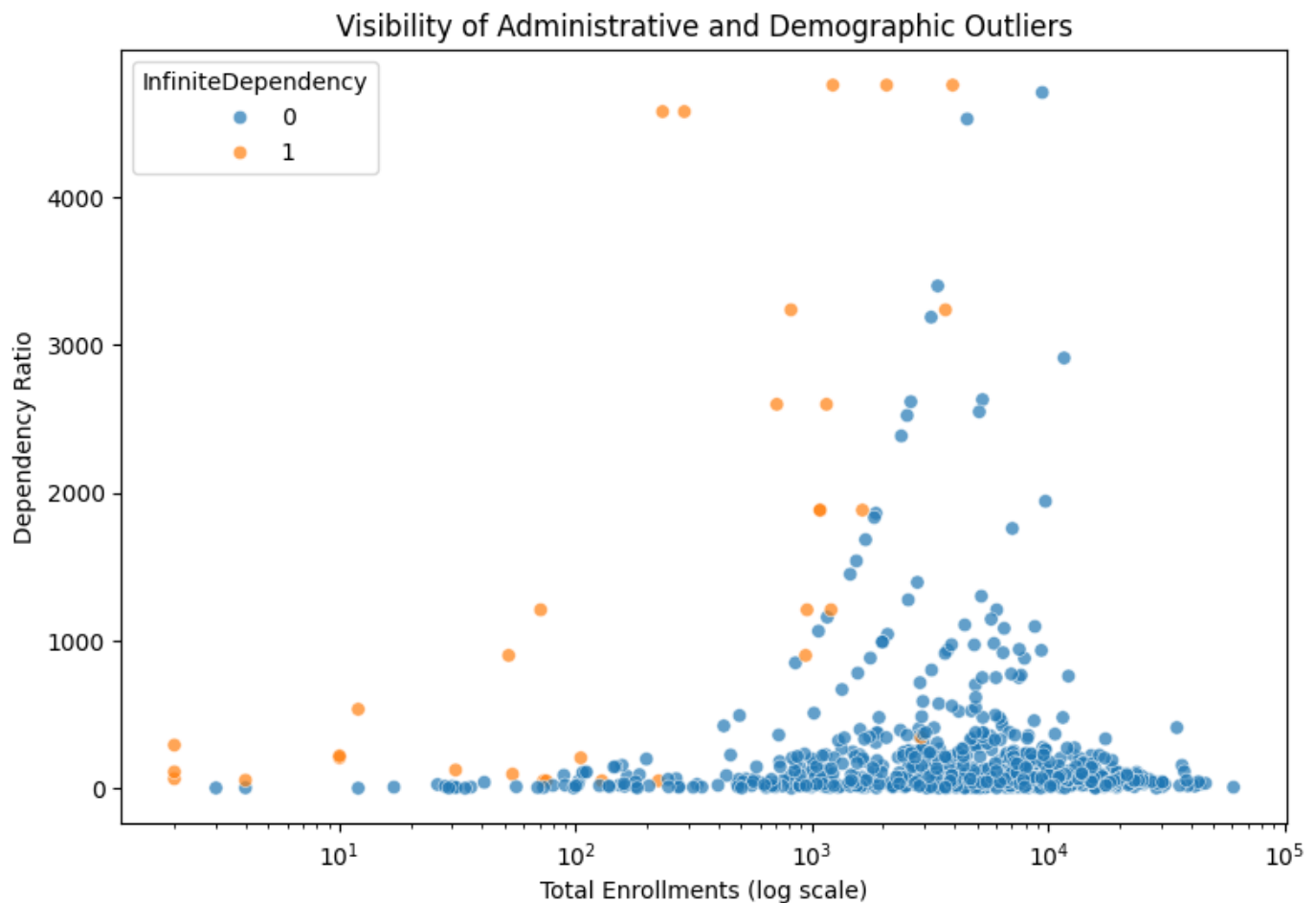
sns.scatterplot(
    data=uidai_clean,
    x="total_enrolled",
    y="dependency_ratio",
    hue="InfiniteDependency",
    alpha=0.7
)

plt.xscale("log")

plt.xlabel("Total Enrollments (log scale)")
plt.ylabel("Dependency Ratio")

plt.title("Visibility of Administrative and Demographic Outliers")

plt.show()
```



Visibility of Administrative and Demographic Outliers

- The log-scaled enrollment axis shows that Aadhaar enrollment activity spans across districts of vastly different population scales, ranging from very small administrative units to extremely large enrollment ecosystems.

- Districts marked as **InfiniteDependency = 1** (orange points) represent regions where adult enrollments were recorded as zero, creating structurally infinite dependency conditions.

- These districts are not indicative of incorrect data or administrative failure.

Instead, they highlight:

- extreme demographic imbalance,
- absence of adult enrollment participation,
- or structurally constrained enrollment ecosystems.

- To preserve analytical continuity, these districts were dynamically assigned dependency ratios based on:

- **(Maximum State Dependency Ratio + 50)**

This allows them to remain visible as structural outliers without introducing undefined or missing values into the analysis.

Clear Separation of Structural Outliers

The visualization shows that infinite dependency districts form visibly isolated clusters at extremely high dependency levels.

This separation helps:

- distinguish genuine structural anomalies from normal demographic variation,
- prevent distortion of overall enrollment maturity analysis,
- and preserve visibility of severe demographic stress zones.

High Enrollment Does Not Eliminate Structural Risk

Several infinite dependency districts still appear at:

- moderate or large enrollment volumes

This demonstrates that:

- high enrollment activity alone cannot guarantee demographic maturity,
- because dependency imbalance may still dominate the enrollment structure.

Administrative and Policy Relevance

This visualization enables policymakers to identify districts requiring:

- special demographic interpretation,
- custom enrollment strategies,
- and long-term structural monitoring rather than routine performance evaluation.

The chart therefore acts as a structural anomaly visibility map rather than a conventional performance comparison plot.

In []:

